

Collaborative Research: Assessing Wild Bees' Resilience to Heat Waves across Scales of Biological Organization

Overview:

Insects account for most of the world's animal species and provide critical ecosystem services, yet surprising research gaps remain in our understanding of how insects are being affected by the increasing frequency of extreme weather events. In the proposed work we will bridge the individual, population, and community scales to determine how short-term extreme heat events (heat waves) impact native bees across their different life history stages. First, we will use experimentally simulated heat waves on a model bee species, *Osmia lignaria*, to measure behavior and survival across life stages. Second, we will track marked adult *O. lignaria* at field sites in central California to measure how heat waves impact behavior, survival, and reproduction in the wild. Third, we will use AI-assisted camera traps to measure the activity of all bee species before, during, and after naturally occurring heat waves; to our knowledge this will be the first large-scale study of how heat waves are impacting native bee species in the wild. By integrating the effects of heat waves from the individual to the community level, across the adult, egg, and larval life stages, and across spatial scales from greenhouse experiments to landscapes, this study will provide a comprehensive assessment of how heat waves impact wild bees.

Intellectual merit:

This research will explore the idea that resilience (that is, the capacity of a system to deal with change and continue to develop) increases with increasing scale of biological organization and ecological complexity. We expect to find that bees show more resilience to heat waves as we move from single-species lab experiments to observations of multi-species bee communities in the wild. Specifically, we predict low resilience in simulated heat wave experiments, where immobile bee larvae developing in nests may exceed their critical thermal limits. We predict increased resilience in the field, because bees select cooler nesting microhabitats that protect developing offspring from heat waves that occur later in the season. When we measure the response of all bee species to naturally occurring heat waves, we expect to find differential heat tolerance among bee taxa, which could make the response at the community level more resilient as compared with the single-species case. By bridging from single-species greenhouse experiments to observations of many bee species over hundreds of square kilometers, this research will reveal at what scale(s) resilience mechanisms emerge and buffer bee populations to extreme temperatures. The research will also provide some of the first 'big data' on insect responses to heat waves in nature. We will develop the efficacy of two newly-available technologies – automated remote cameras and AI-assisted insect identification – which promise to make data collection on insects vastly more efficient.

Broader impacts:

The limited understanding of how extreme weather affects pollinators, along with the absence of scalable monitoring tools, currently constrains pollinator conservation. Although heat waves are widely suspected to contribute to pollinator declines, there are no large-scale datasets documenting their impacts on bees; this project will generate the first such dataset. The work will also address a major barrier to long-term monitoring of North American native bees: the lack of non-lethal, scalable identification methods that do not require extensive taxonomic expertise. The monitoring technologies that we build for this project could transform monitoring efforts currently pursued by agencies such as the Bureau of Land Management, National Park Service, and United States Fish and Wildlife Service, offering substantial cost savings relative to technician-based surveys. The PIs' partnerships with agencies and conservation NGOs—including the National Native Bee Monitoring Network—will facilitate translation of results into practice. To advance collaboration, we will organize the first national Automated Insect Monitoring workshop, bringing together technology developers, researchers, and government agencies for knowledge transfer, demonstrations, training, and research exchange at the Bell Museum of Natural History. In partnership with the museum's Public Engagement and Science Learning team, we will also develop public programs—including an interactive camera-trap exhibit, participation in public science events, and a middle-school summer camp module on wild bees—to engage audiences in entomology and monitoring science, while evaluating impacts through visitor surveys and participant feedback.